

29 Oct 2021

EX1

$$V_{B1} = V_{cc} \frac{R_{B2}}{R_{B1} + R_{B2}} = 1.091 \text{ V}$$

$$R_{BB} = R_{B1} || R_{B2} = 10.91 \text{ k}\Omega$$

$$V_{B1} - I_{B1} R_{BB} - V_{BE1} - I_{C1} (1 + \frac{1}{\beta_F}) R_{E1} = 0$$

$$\frac{V_{B1} - V_{BE1}}{R_{BB} + (1 + \frac{1}{\beta_F}) R_{E1}} = I_{C1} = 0.349 \text{ mA}$$

$$R_C (I_{C1} - I_{C2}) = R_{E2} (I_{C2} (1 + \frac{1}{\beta_F}) - V_{BE2})$$

$$\frac{R_C I_{C1} + V_{BE2}}{\frac{R_C}{\beta_F} + R_{E2} (1 + \frac{1}{\beta_F})} = I_{C2} = 0.565 \text{ mA}$$

$$V_{CE1} = V_{cc} - R_C (I_{C1} - I_{C2}) - R_{E1} I_{C1} (1 + \frac{1}{\beta_F}) = 4.094 \text{ V}$$

$$V_{CE2} = -(V_{cc} - R_{E2} I_{C2} (1 + \frac{1}{\beta_F})) = R_{E2} I_{C2} (1 + \frac{1}{\beta_F}) - V_{cc} = -5.152 \text{ V}$$

$$Q_1 (I_{C1}, V_{CE1}) = (0.349 \text{ mA}, 4.094 \text{ V})$$

$$Q_2 (I_{C2}, V_{CE2}) = (0.565 \text{ mA}, -5.152 \text{ V})$$

small signal: topology of feedback

current mixing (shunt)

voltage sampling

$\frac{V_o}{V_s}$ = transconductance amplifier

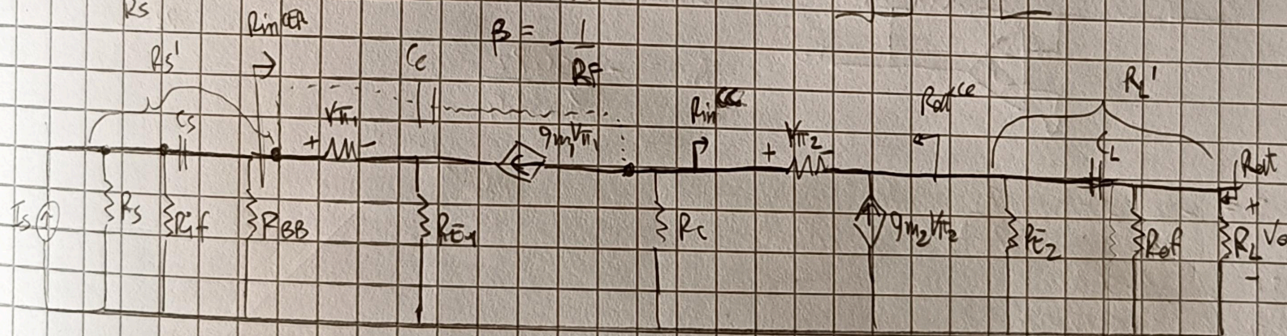
Norton eq.

$$I_s = \frac{V_s}{R_s}$$

$$R_{if} = R_F$$

$$R_{of} = R_F$$

$$\beta = -\frac{1}{R_F}$$



$$A_a = \frac{V_o}{V_s} \Big|_{\beta=0}$$

$$\alpha_{cc} = \frac{(1 + \beta_0) R_L}{R_{in,cc}} = 0.992 \text{ [V/V]}$$

$$g_{m1} = \frac{I_{C1}}{V_T} = 0.01396 \text{ S}$$

$$R_L' = R_{E2} || R_{of} || R_L = 5.26 \text{ k}\Omega$$

$$\alpha_{CER} = -\frac{\beta_0 (R_C || R_{in,cc})}{R_{in,CER}} = -19.53 \text{ [V/V]}$$

$$g_{m2} = \frac{I_{C2}}{V_T} = 0.0226 \text{ S}$$

$$R_s' = R_s || R_{if} || R_{BB} = 7.77 \text{ k}\Omega$$

$$\alpha_i = R_s' || R_{in,CER} = 7249 \text{ }\Omega$$

$$r_{\pi 1} = \frac{\beta_0}{g_{m1}} = 7.163 \text{ k}\Omega$$

$$R_{in,cc} = r_{\pi 2} + (1 + \beta_0) R_L' = 535.7 \text{ k}\Omega$$

$$A_a = \alpha_i \alpha_{CER} \alpha_{cc} = -140.44 \text{ V/V}$$

$$r_{\pi 2} = \frac{\beta_0}{g_{m2}} = 4.424 \text{ k}\Omega$$

$$R_{in,CER} = r_{\pi 1} + (1 + \beta_0) R_{E1} = 108.2 \text{ k}\Omega$$

$$A_F = \frac{A_a}{1 + \beta_0 A_a} = -72.53 \text{ k}\Omega$$

$$A_V = \frac{V_o}{V_s} = A_F = -2.198 \text{ [V/V]}$$

$$R_{out} = R_L' \parallel R_{out} = 248 \Omega$$

$$R_{out} = \frac{R_{T2} + R_L}{1 + \beta_0} = 260 \Omega$$

$$R_{out}^F = \frac{R_{out}}{1 + \beta_{out}} = R_{out} = 128 \Omega \quad \checkmark$$

$$C_c = 2 pF \rightarrow f_H$$

applying Miller

$$G = A_{v,CE} = -19.53 \text{ [V/V]}$$

$$C_{c1} = C_c (1 - G) = 41.06 pF$$

$$C_{c2} = C_c (1 - \frac{1}{G}) = 2.102 pF$$

C_s, C_c are shorted at high frequency

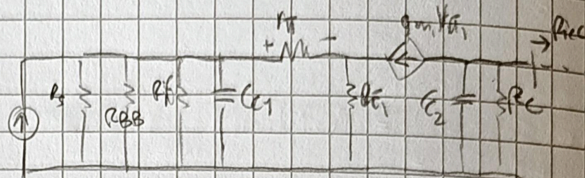
$$R_{cc1} = R_s' \parallel R_{in,CE} = 7249 \Omega$$

$$R_{cc2} = R_c \parallel R_{in,CC} = 2132 \Omega$$

applying OCTC method

$$f_H^2 = \frac{1}{2\pi} \left(\frac{1}{C_{c1} R_{cc1} + C_{c2} R_{cc2}} \right) = \frac{1}{2\pi} \frac{1}{2.98 \cdot 10^{-7} + 4.44 \cdot 10^{-8}} = 465.28 \text{ kHz}$$

$$f_H^F = f_H^2 (1 + \beta_{out}) = 900.9 \text{ kHz}$$



Ex 2

$$R_1 = 1 k\Omega$$

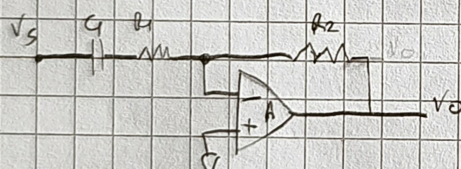
$$R_2 = 100 k\Omega$$

$$G = 1 mF$$

$$A(s) = \frac{10^5}{1 + \frac{s}{10^2}} \text{ [V/V]}$$

$$\omega_0 = 100 \text{ rad/s}$$

$$A_{db} = 10^5$$



$$1) \text{ inverting configuration } A_v = -\frac{Z_2}{Z_1} = -\frac{R_2}{R_1 + \frac{1}{sC}}$$

$$A_v = -\frac{sCR_2}{1 + sCR_1}$$

$$2) \text{ P.T.: } T = \frac{-x_r}{x_s} = A_d(s) \cdot \frac{Z_1}{Z_1 + Z_2} \quad \text{direct loop inspection}$$

$$\beta = \frac{Z_1}{Z_1 + Z_2} = \frac{1 + sCR_1}{1 + sC(R_1 + R_2)}$$

$$\omega_p = \frac{1}{CR_1 + CR_2} = 9.9 \cdot 10^3 \text{ rad/s}$$

$$\beta \neq 1 \rightarrow \frac{1}{\beta} = 0 \text{ dB}$$

$$\omega_z = \frac{1}{CR_1} = 10^6 \text{ rad/s}$$

$$\beta \neq 1 \rightarrow \frac{1}{\beta} = 0 \text{ dB}$$

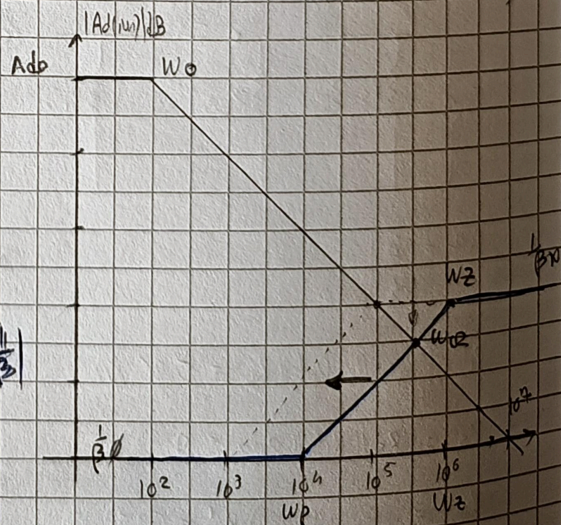
$$\frac{1}{\beta} = 1 + \frac{R_2}{R_1} = 40 \text{ dB}$$

$$\omega_{cr} = \left\{ \begin{array}{l} A_{db} \omega_0 = A_x \omega_{cr} \\ \frac{1}{\beta_0} \omega_p = \frac{A_x}{\omega_{cr}} \end{array} \right\} \quad A_x = \frac{\omega_{cr}}{\beta_0 \omega_p}$$

$$A_{db} \omega_0 \beta_0 \omega_p = \omega_{cr} = 3.15 \cdot 10^5 \text{ rad/s}$$

$$PM = 180^\circ - \arg\left(\frac{\omega_{cr}}{\omega_0}\right) - \arg\left(\frac{\omega_{cr}}{\omega_p}\right) + \arg\left(\frac{\omega_{cr}}{\omega_z}\right)$$

$$= 19.3^\circ \text{ low phase margin}$$



$\phi = R_1$ so that $PM = 45^\circ$

exactly $PM = 45^\circ \rightarrow 180 - \overset{90^\circ}{\cancel{\arg\left(\frac{W_R}{W_0}\right)}} - \arg\left(\frac{W_p}{W_0}\right) + \arg\left(\frac{W_z}{W_0}\right) = 45^\circ$

$45^\circ - \arg\left(\frac{W_R}{W_p}\right) + \arg\left(\frac{W_z}{W_z'}\right) = 0$

$AdW_0 = W_R \cdot \frac{1}{\beta}$

$W_R = AdW_0 \beta = AdW_0 \left(\frac{R_1}{R_1 + R_2} \right)$

$45^\circ - \arg\left[\frac{AdW_0 \left(\frac{R_1}{R_1 + R_2} \right)}{\frac{1}{G(R_1 + R_2)}} \right] + \arg\left[\frac{AdW_0 \left(\frac{R_1}{R_1 + R_2} \right)}{\frac{1}{G(R_1)}} \right] = 0$

$R_1 = 3109 \Omega$

$W_R = 3.05 \cdot 10^5 \text{ rad/s}$

$W_p = 9.7 \cdot 10^3 \text{ rad/s}$

$W_z = 3.216 \cdot 10^5 \text{ rad/s}$

$PM = 180 - \arg\left(\frac{W_R}{W_0}\right) - \arg\left(\frac{W_p}{W_p'}\right) + \arg\left(\frac{W_z}{W_z'}\right) = 45.01^\circ$

for an approximate estimate we may consider that $W_z = W_R \rightarrow R_1' = \frac{1}{W_R G} = 3174 \Omega$

$W_R = 3.15 \cdot 10^5 \text{ rad/s}$

$W_z = 3.15 \cdot 10^5 \text{ rad/s}$

$W_p = 9.652 \cdot 10^3 \text{ rad/s}$

$PM = 46.8^\circ$

4) gain at 10^7 rad/s

high freq

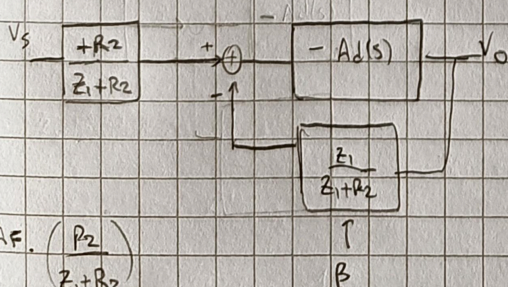
$A_{ol} = -Ad(s) \approx -1$

$\frac{1}{\beta} = 32 \text{ dB} = 32$
 $W = 10^7 \text{ rad/s}$

$T = \frac{-X_1}{X_s} \rightarrow T = \beta A_{ol} = \frac{1}{32} \ll 1$

$A_F \approx A_{ol} = -1$

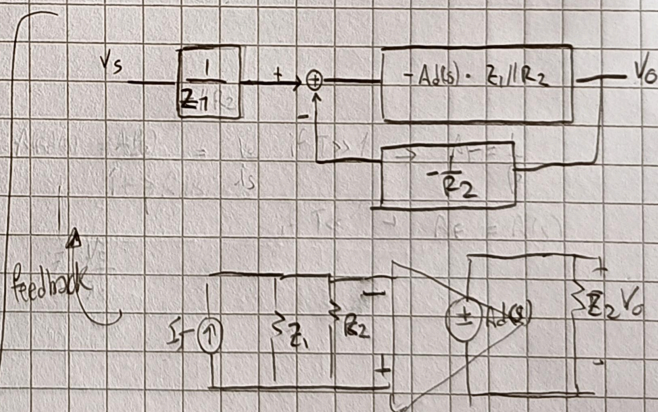
direct loop



$\frac{V_0}{V_s} = A_F \cdot \left(\frac{R_2}{Z_1 + R_2} \right) = -1 \cdot (0.96) = -0.96$

on $R_1 = 3109 \Omega$

$\neq A_v = \frac{-s R_2}{1 + s R_1} \text{ ? Why?}$



direct loop inspection

1 ora e 20 minuti